

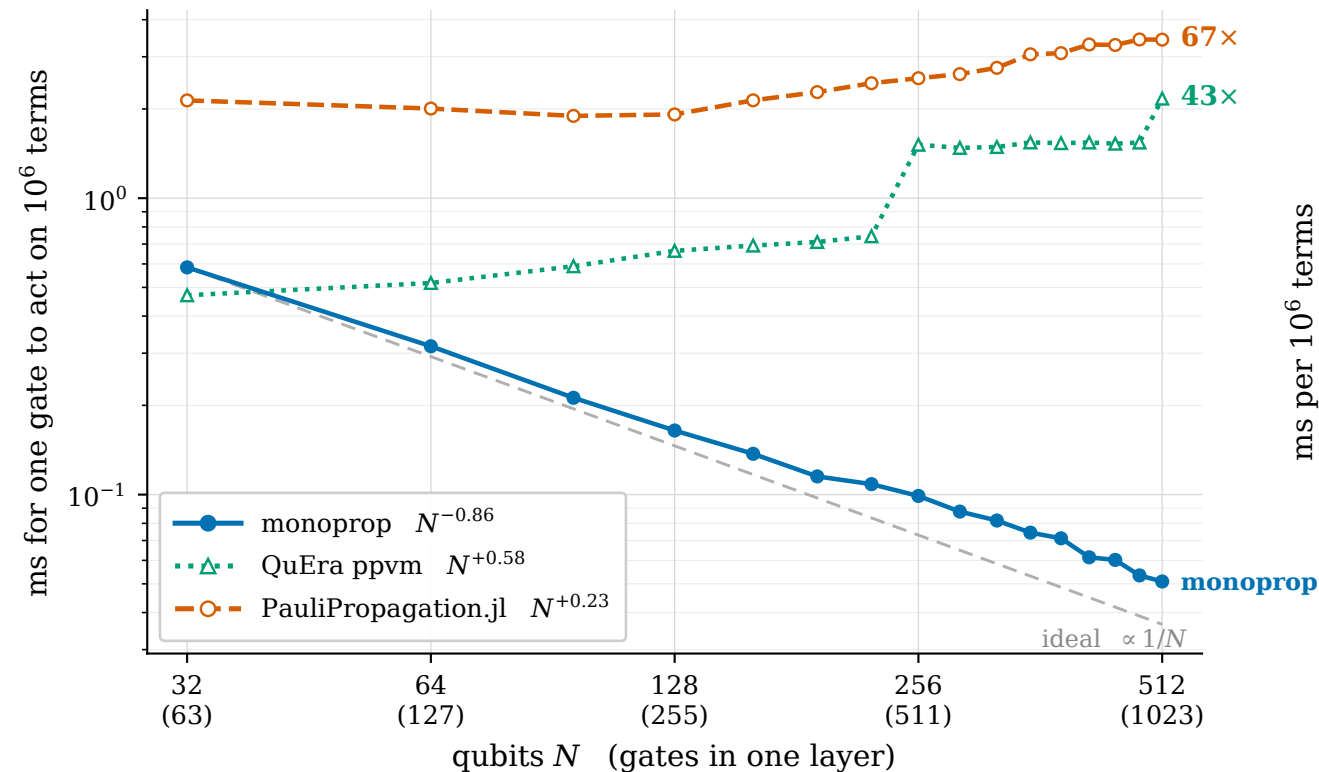
One gate, one million terms: monoprop's cost per gate falls as the system grows

kicked-Ising chain, N qubits, open boundary · 5 layers of $[R_{zz}(\pi/4)$ on every bond, then $R_x(\pi/4)$ on every site] · observable $\sum_i Z_i$, Heisenberg picture

Pauli-weight cutoff 6, no coefficient pruning · 119,280 → 2,310,000 terms in (a), 119,280 fixed in (b) · one thread, one host

all three engines apply the same circuit and agree on every term count and on $\langle O \rangle$ to 12 digits

(a) the marginal cost of a gate falls as $1/N$



(b) why: monoprop hardly notices the extra width

